

HARDWARE

TASK: 1

Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

Steps to Connect and Test a USB Webcam

1. **Connect the webcam:** Plug the USB webcam into an available USB port on the computer.
2. **Wait for detection:** Windows should automatically detect the webcam and install the required drivers. If a driver is needed, install it using the driver software provided with the webcam.
3. **Check the webcam:** Open **Settings** → **Bluetooth & devices** → **Cameras** and make sure the webcam appears in the list.
4. **Open Camera app:** Search for **Camera** in the Windows Start menu and open the default Camera application.
5. **Select the webcam:** If required, select the connected USB webcam as the camera source.
6. **Test the video:** Check that a live video preview appears and that the image is clear.
7. **Capture a photo:** Click the **Take Photo** button in the Camera app.
8. **Verify the photo:** Open the saved image to confirm that the webcam successfully captured the photo.

Result: The USB webcam was successfully connected, the required driver was installed/detected, and the webcam was verified by capturing a photo using the Windows Camera app.

TASK: 2

Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Steps to Set Up and Test a Local Printer

1. **Connect the printer:** Connect the printer to the computer using a USB cable, or connect both the printer and computer to the same Wi-Fi network for a network printer.
2. **Turn on the printer:** Switch on the printer and make sure it has paper and sufficient ink/toner.
3. **Install the driver:** Windows may automatically install the correct driver. If not, download and install the printer driver for the exact printer model from the manufacturer's website.
4. **Add the printer:** Go to **Settings** → **Bluetooth & devices** → **Printers & scanners** → **Add device**.
5. **Select the printer:** Choose the connected printer and complete the installation.
6. **Set as default:** If needed, select the printer and choose **Set as default**.
7. **Print a test page:** Open **Printers & scanners**, select the printer, choose **Printer properties**, and click **Print Test Page**.
8. **Verify the result:** Check that the test page prints correctly with clear text and images.

Result: The local printer was successfully connected, the correct driver was installed, and a test page was printed successfully, confirming that the printer is working.

TASK: 3

Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

Simulate & Resolve “Printer Not Detected” Problem Using Device Manager

1. Disconnect the printer



Turn off the printer and unplug its USB cable from the computer.

2. Check the problem



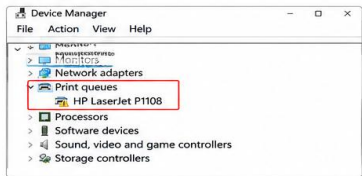
Open Settings > Printers & scanners and confirm the printer is not available.

3. Open Device Manager



Right-click Start button and select Device Manager.

4. Check for errors



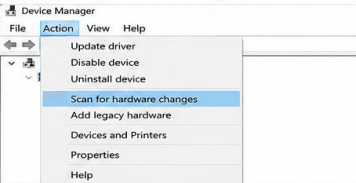
Look under Printers, Print queues, or USB controllers for yellow warning icon.

5. Reconnect the printer



Plug the USB cable back into the computer and turn on the printer.

6. Scan for hardware changes



In Device Manager, click Action > Scan for hardware changes.

7. Update driver if required



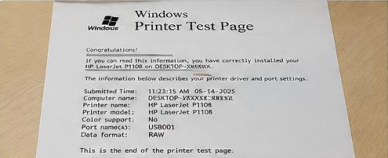
If the printer still has an error icon, right-click it and select Update driver > Search automatically for drivers.

8. Verify the printer



Go back to Settings > Printers & scanners and check that the printer appears.

9. Test printing



Print a test page to confirm that the printer is working correctly.

 **Result:** The printer was successfully detected again after reconnecting the USB cable and using Scan for hardware changes in Device Manager. The printer can now be used normally.

Steps to Simulate and Resolve a “Printer Not Detected” Problem

- 1. Disconnect the printer:** Turn off the printer and unplug its USB cable from the computer.
- 2. Check the problem:** Open Settings → Bluetooth & devices → Printers & scanners and confirm that the printer is not available or shows as disconnected.
- 3. Open Device Manager:** Right-click the Start button and select Device Manager.

4. **Check for errors:** Look under **Printers, Print queues, or Universal Serial Bus controllers** for the printer. Look for a **yellow warning triangle** or other error icon.
5. **Reconnect the printer:** Plug the USB cable back into the computer and turn on the printer.
6. **Scan for hardware changes:** In Device Manager, click **Action** → **Scan for hardware changes**. Windows should detect the connected printer again.
7. **Update the driver if required:** If the printer still has an error icon, right-click it and select **Update driver** → **Search automatically for drivers**.
8. **Verify the printer:** Return to **Settings** → **Bluetooth & devices** → **Printers & scanners** and check that the printer appears.
9. **Test printing:** Print a test page to confirm that the connection and driver are working correctly.

Result: The printer was successfully detected again after reconnecting the USB cable and using **Scan for hardware changes** in Device Manager. The printer can now be used normally.

TASK: 4

A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

Steps to Troubleshoot and Resolve

1. **Close other camera applications**
 - Close the **Camera app**, Microsoft Teams, Skype, WhatsApp, or any other application that may be using the webcam.
 - Restart Zoom and check the camera again.
2. **Check Zoom camera settings**
 - Open **Zoom** and sign in.
 - Go to **Settings** → **Video**.
 - Under **Camera**, select the correct USB webcam from the drop-down list.
 - Check whether the video preview appears.
3. **Check Windows camera permissions**
 - Open **Settings** → **Privacy & security** → **Camera**.
 - Turn on **Camera access**.
 - Turn on **Let apps access your camera**.
 - If available, make sure **Let desktop apps access your camera** is also enabled because Zoom is a desktop application.
4. **Restart Zoom**
 - Completely close Zoom.
 - If necessary, open **Task Manager** with **Ctrl + Shift + Esc** and end any remaining Zoom processes.
 - Open Zoom again and test the camera.
5. **Check the webcam in Device Manager**
 - Right-click **Start** → **Device Manager**.
 - Expand **Cameras** or **Imaging devices**.
 - Find the webcam and check for a yellow warning icon.
 - If there is an error, right-click the webcam and select **Update driver**.
6. **Reconnect the webcam**

- Disconnect the USB webcam.
 - Wait a few seconds and connect it to another USB port.
 - Open Zoom again and select the webcam under **Settings → Video**.
7. **Test in a Zoom meeting**
- Start a test meeting or join a meeting.
 - Click the **arrow next to the camera icon**.
 - Select the correct webcam and confirm that the video is displayed.

Result

The webcam should now work in both the **Windows Camera app and Zoom**. If it works in Camera but not Zoom, the most common solutions are **selecting the correct camera in Zoom, enabling Windows camera permissions, and closing other applications that are using the webcam**.